GLOBAL ENERGY HACKATHON 2026 · INNOVATIVE & AFFORDABLE SOLUTIONS

GridRescue AI

The problem, with evidence: hospitals on weak grids keep their theatres alive on diesel, and a hospital that moves cannot take rooftop solar with it.

THE MARKET · PUBLISHED FIGURES, SOURCE ON EACH

About half
of hospitals in sub-Saharan Africa have reliable electricity
WHO · World Bank · IRENA · SEforALL, Energizing health

≈ 60 % · ≈ 1 %
of the world's best solar resources are in Africa — with about 1 % of installed solar capacity
IEA, Africa Energy Outlook

₦3,277 / L
diesel, up 86 % in a year — national average, not an Enugu pump price
NBS Automotive Gas Oil price watch

Most firms in sub-Saharan Africa report power outages (World Bank Enterprise Surveys). Five outage challenges the platform addresses: grid interruptions and load shedding · transmission and distribution failures · severe weather and infrastructure damage · diesel shortages, cost and generator failures · cyber or operational disruptions (planned scenario).

WORKED EXAMPLE · NATIONAL ORTHOPAEDIC HOSPITAL, ENUGU · 350 BEDS (REPORTED)

≈ 6 h
of grid a day — head engineer (reported)

1,550 kW
connected; unstable supply hits equipment (reported)

≈ 5 years
until the move to a new building (brief)

10–30 s
to the generator at every grid drop (reported)

BRIEF
12 · Relocatable solar for a hospital on the move

TRACK
Clean Energy Solutions

TEAM
Secure AI Grid · Pavan Manjunath · Thomas Prüfer
Pavan: product, AI and engineering · Thomas: energy systems and the business case

HOW THIS PITCH IS SCORED

Impact	25%
Innovation	25%
Sustainability	25%
Feasibility	15%
Scalability	10%

Access · Affordability · Availability "realistic to implement within 6–18 months · adaptable across different global contexts"

From a map pin to an investment-ready plan

GridRescue AI is vendor-neutral decision-support software: pin a hospital on the map and it reads the site from open data, simulates every outage hour by hour, designs the solar + energy system + UPS + generator stack, prices it, plans the move and writes the ESG report — in minutes, for any pin on the map.

BUILT DURING THE GLOBAL ENERGY HACKATHON 2026 · FOUR MODULES, ONE FLOW

1 · GRID MAPPER + NEW-SITE LENS

Grid strength within 50 and 100 km, supply hours and tariff from open data; the lens ranks new sites within 12 km.

2 · RELOCATION PLAN

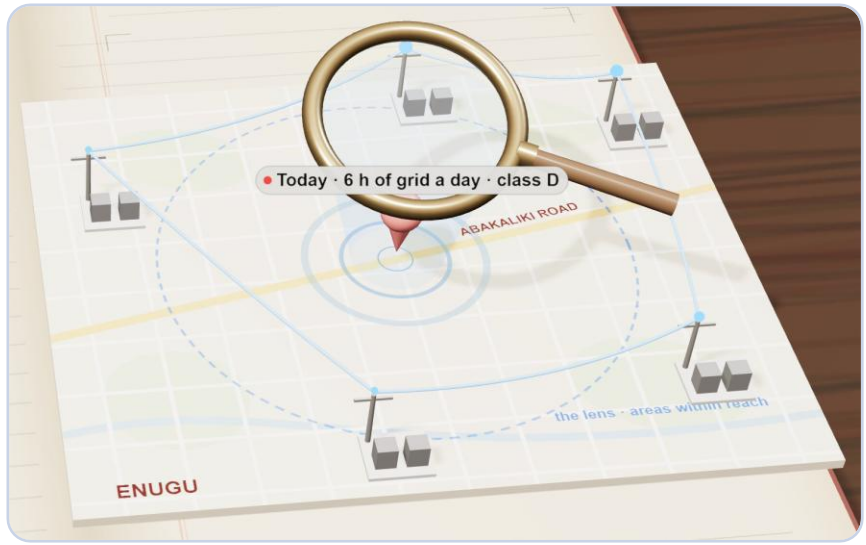
The brief's five questions answered: two sites, theatres-first loads, move plan, 5- and 10-year money, funders.

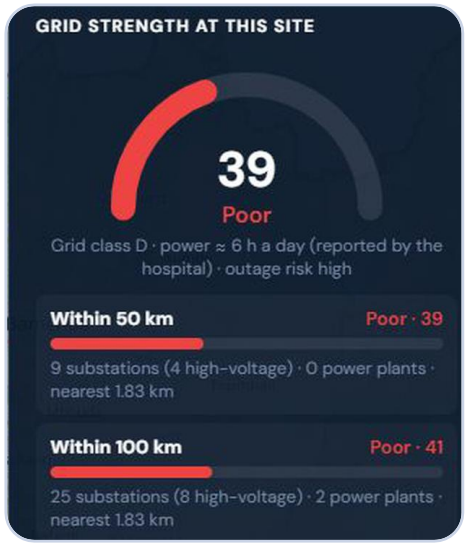
3 · 3D OUTAGE SIMULATION + ML PICK

Hour by hour, four stress tests; Random Forest pick of the energy system (80.4 % agreement), verified by the simulation.

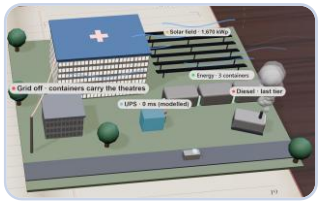
4 · ESG REPORT

Scope 1 and Scope 2, E·S·G lines, carbon value — written only after the simulation verifies the design.





Unique: the grid mapper reads grid strength within 50 and 100 km and the "hospital is moving" lens ranks the new site.

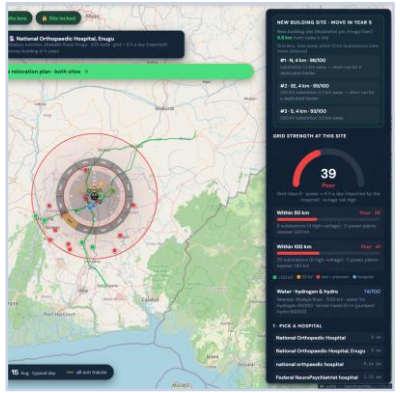


Three layers: online UPS on theatres and ICU, 0 ms (modelled) · solar by day, energy containers by night · generator last.

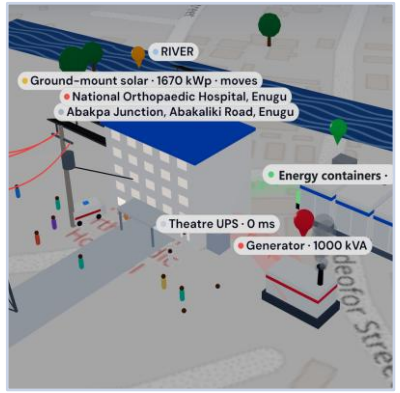
Three A's: Access: any site in seconds · Affordability: ₦0 upfront, from the diesel budget · Availability: power that moves.

User · who pays: hospital engineering teams; GridRescue AI earns software fees, not the energy fee.

The prototype, running on the worked example



Grid mapper: strength 39 / Poor; the new-site lens ranks areas for the move



3D scene: solar that moves, containers, theatre UPS, generator last

BUILT DURING THE GLOBAL ENERGY HACKATHON 2026

Measured or published inputs: the hospital's reported supply hours, the NBS diesel price, the EERC tariff, four open sun sources, OpenStreetMap grid data. **Modelled outputs:** everything the engine computes — ±30 %, to be confirmed with the hospital's load data.

PIN DROP → SITE READ → SIMULATION → DESIGN → COST → PAYBACK → REPORT

1	**PIN DROP**	National Orthopaedic Hospital, Enugu, Abakpa Junction — any pin on the map runs the same flow	INPUT
2	**SITE READ**	≈ 6 h grid a day (reported) · grid class D “Poor”, 9 substations within 50 km (OpenStreetMap, modelled) · 5.48 peak-sun hours (four open sources) · ₦160/kWh EERC Band A · diesel ₦3,277/L (published)	REPORTED · PUBLISHED
3	**OUTAGE SIMULATION**	Hour by hour in 3D: 4 of 4 stress tests pass (12 h evening outage, 24 h grid collapse, worst solar month, 18 h); 100 % of the critical load in the modelled outages; any outage up to 24 h survived	MODELLED
4	**RECOMMENDED DESIGN**	1,670 kWp ground-mounted solar · 5,020 kWh containerised energy system (8.7 h) · 500 kVA online UPS, 0 ms (modelled) · existing 1,000 kVA generator last · ML pick 76 %, verified by the simulation	MODELLED
5	**COST**	Capex ₦3.85 bn bought outright; as a service ₦69.1 M a month, ₦0 upfront, against ₦299.2 M a month of modelled diesel for the protected load	MODELLED
6	**PAYBACK**	Bought outright: diesel savings repay the system and the move in 1.6 years. 10 years, all energy costs of the protected load: ₦37.6 bn → ₦15.9 bn	MODELLED
7	**REPORT**	ESG: 2,246 t CO_2 a year avoided, net (Scope 1 − Scope 2) · move plan: lens area #1, 4 km north, about three weeks, ₦678 M, 88 % of the investment reused	MODELLED

Judges look for: a working prototype, shown not claimed — every input above is labelled reported or published, every output modelled (engine results, ±30 %, to be confirmed with the hospital's load data).

Theatres on in every modelled outage

KPI (engine result, modelled)	Today (diesel)	With the system	What changed
Theatre-hours a year without grid (4 suites, 12,480 a year)	4,992 (40 %)	0	**Theatres on in every modelled outage**
Generator switchovers reaching equipment, a year	≈ 1,460	0 (online UPS; 0 ms, modelled)	**No 10–30 s gap mid-surgery**
Diesel for the protected load, a year	≈ 1,042,790 L	≈ 176,000 L	**866,678 L avoided (83 %)**
Operating cost of the protected load, a month	₦299.2 M modelled diesel	₦69.1 M service fee, ₦0 upfront	**Saves whenever real diesel > ₦117.9 M**
CO_2 a year (2.68 kg per litre, IPCC default)	2,323 t Scope 1 (diesel)	77 t Scope 2 grid charging	**2,246 t avoided, net**

WHO FUNDS THE HARDWARE

Rural Electrification Agency public-sector solarisation programme (1.5 MW solar hybrid at the National Orthopaedic Hospital, Dala, Kano) · EU and Enugu State ₦4.3 bn Solar for Health programme (25 facilities) · development-bank financing for health-facility electrification · ESCOs with concessional debt: the ESCO owns, maintains and moves the kit; the hospital pays a service fee from its diesel budget.

THREE MONEY STORIES, NEVER MIXED

(a) the hospital's savings: diesel avoided minus service costs (year one ₦3,764.5 M → ₦1,588.9 M of energy costs, engine); simple payback applies only when the hospital buys outright. **(b)** the ESCO's project cash flow: fee, capex, O&M, relocation clause, concessional debt — the ₦69.1 M a month is this story, not our revenue. **(c)** GridRescue AI's software revenue minus operating costs. All figures are engine results, ±30 %, to be confirmed with the hospital's load data; the real diesel bill is partner data still owed.

1.6 years

Bought outright: diesel savings repay the system and the move (modelled)

₦21.8 bn saved

10 years, all energy costs of the protected load: ₦37.6 bn → ₦15.9 bn (modelled)

88 % reused

of the investment after the move: 4 km north, about three weeks, ₦678 M (modelled)

Judges look for: a large, quantified benefit on the brief's own metric, a solution that pays for itself and lowers environmental impact — every figure here is an engine result (modelled, ±30 %) for the worked example, with the funders named.

Why fund this software company

CUSTOMER SEGMENTS

Hospitals and clinics on weak grids (first: federal and teaching hospitals, Nigeria) · Healthcare networks and health programmes · ESCOs, energy-system integrators, hospital energy contractors · Utilities (feeder planning) · Later: data centres, water, telecom, industrial and agricultural sites.

VALUE PROPOSITION

One workflow from a map pin to an investment-ready plan: site read from open data → hour-by-hour outage simulation → vendor-neutral solar + energy system + generator design → cost, payback and ESG report · Transparent assumptions, no hardware sales.

REVENUE STREAMS

Paid site assessment (US$ 2,000 per site, proposed) · Annual software licence (US$ 3,000 per site, proposed) · Enterprise multi-site and utility licences · Integration / white-label fees · Later: performance-based fee after verified savings.

THE ASK · US$ 150,000 SEED IN TWO TRANCHES

Use of funds (US$): Software engineering and productionisation **70,000** · Energy engineering and model validation **20,000** · Hospital pilot integration and field validation **15,000** · Security, testing and hardening **12,000** · Customer acquisition and partner onboarding **10,000** · Contingency **10,000** · Cloud hosting, weather and geospatial data **8,000** · Legal, contracts and IP **5,000** · Total **150,000**

Tranche 2 (US$ 75,000) after the engine passes agreed validation tests against hospital data and a pilot partner is secured. Software only; hospital hardware is project-financed separately.

Roadmap 6–18 months: 0–3 validate the engine on the hospital's load audit and generator logs, security hardening · 3–9 two paid pilots (Enugu first) by month 9, with an ESCO and the local distribution company · 9–12 commercial release, 10 paying sites by month 12 · 12–18 utilities and other critical facilities, further outage scenarios (weather groups, cyber/operational contingencies), new countries by adding a tariff table — any pin on the map works.

WHY PAY WHEN REOPT IS FREE

REopt (NREL, free): sizes PV, energy systems and a generator so a site survives a specified outage; built for energy experts.

HOMER Pro (UL Solutions, commercial): annual microgrid simulation and least-cost design; priced for consultants.

HOMER Powering Health Tool (World Bank/ESMAP, free): a sizing and cost tool, not an outage-by-outage simulation.

Odyssey Energy Solutions: project planning, financing, procurement and monitoring for developers.

MapYourGrid / OpenStreetMap: our data partner, not a competitor.

WHAT NONE OF THEM DOES IN ONE FLOW — GRIDRESCUE AI DOES TODAY

1 From a map pin to an investment-ready hospital plan in minutes, on open data, with no API keys and no expert.

2 The grid mapper with the “hospital is moving” new-site lens: substations and lines within 50 and 100 km, published supply hours, areas ranked for the new building — then a relocation plan with a move cost and a reuse percentage.

3 Every energy-system option simulated hour by hour on the site's own sun, grid hours and load, four stress tests, with a vendor-neutral ML classification of the energy system that the simulation verifies.

4 Local market assumptions built in: the NBS diesel price, the state tariff, Nigerian funders and programmes.

5 The ESG report (Scope 1 and Scope 2, E·S·G lines) and the before → after KPIs written before anything is bought.

GridRescue AI turns energy planning into investment-ready decisions. We start with hospitals, where power is life, and grow to every critical facility. Map the grid. Keep the theatres on. Move with the hospital.